Simulation-Aided SAR Target Classification via Dual-Branch Reconstruction and Subdomain Alignment

Xiaoling Lv, Xiaolan Qiu, *Senior Member, IEEE*, Wenming Yu, and Feng Xu, *Senior Member, IEEE*

***Abstract*— Convolutional neural networks (CNNs) are widely used in image classification, but such methods often require massive labeled data as learning resources. On one hand, synthetic aperture radar (SAR) image interpretation is difficult, resulting in the lack of large-scale datasets with high-quality labels. On the other hand, CNNs are not explainable enough to provide reliable and trusted application services for SAR target recognition. To solve the above problems, physics-based electromagnetic simulated images are used to alleviate the shortage of real data with annotations, and explainability analysis methods are introduced to explain the basis of network decision-making. To address the domain gap between the simulated and measured data, we propose a novel network integrating dual-branch image reconstruction and subdomain alignment (DBRSA). The network completes the reconstruction of the simulated and measured images through the domain-shared encoder and domain-specific decoders, thereby helping the encoder to learn feature extraction methods independent of labels. In addition, the network aligns the feature vectors of similar targets obtained from different domains according to the real or pseudo labels of the samples, so as to further improve the classification accuracy. The experimental results and model decision analysis results demonstrate that the proposed network can improve the performance reliably by reducing the attention to the background noise and increasing the attention to the shadows and contours of the target, effectively reducing the dependence on the number of sample labels in practical application scenarios.**

***Index Terms*— Convolutional neural networks (CNNs), domain adaptation, explainability, synthetic aperture radar (SAR) target classification.**

I. Introduction

As an active microwave remote sensing imaging system, synthetic aperture radar (SAR) has the ability of all-time and all-weather observation. With the continuous development of SAR imaging technology, SAR automatic target recognition (SAR-ATR) has become an important part of SAR image intelligent interpretation. SAR-ATR mainly consists of three stages: detection, discrimination, and classification. Recently, many excellent algorithms based on deep convolutional neural networks (CNNs) have emerged in SAR-ATR with the rapid development of deep learning [1], [2].

However, applying CNNs to SAR-ATR mainly faces two problems. First, deep CNNs usually require a large amount of labeled data to learn how to effectively extract target features, but SAR image annotating is difficult and costly, resulting in a serious shortage of annotation quantity. Second, the methods based on deep learning have always been regarded as "black boxes," and the internal working mechanism lacks explainability, which makes it difficult to provide reliable and trustworthy applications for SAR-ATR.

To address the challenge of insufficient labeled samples, using simulated SAR images as a substitute is a solution that many researchers are trying. Malmgren-Hansen et al. [3] first discussed the feasibility of transfer learning between the simulated SAR images and real ones. The results verified that pretraining with the simulated data can effectively accelerate the convergence of the model and improve the classification accuracy when the labeled real samples are insufficient. With the development of SAR simulation technology, for some targets, the simulated images and the actual images can be visually indistinguishable. However, it is difficult to directly classify the targets using CNNs trained with simulated images only, which has also been demonstrated in [4].

To improve the performance of the simulation-aided SAR target classification, it is necessary to find a method that can effectively shorten the distance between the simulated target features and the real ones. Exactly, domain adaptation aims to realize the information transfer from the source domain to the target domain, that is, to find a general feature extraction method to classify samples from different domains. These methods can provide some solutions for the task of the simulation-aided SAR target classification. Generally speaking, the existing deep domain adaptation methods can be classified into four types: domain-distribution-based approaches, adversarial-learning-based approaches, reconstruction-based approaches, and generation-based approaches. Domain-distribution-based methods [5], [6], [7], [8] aim to extract

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domain-invariant features by minimizing the distribution difference between two domains. Adversarial-learning-based methods [9], [10], [11], [12], [13] mainly make use of domain discriminators to promote domain confusion through an adversarial training manner. Reconstruction-based methods [14], [15] take image reconstruction as an auxiliary task to suppress information loss. Generation-based approaches [16], [17] typically generate pseudosamples that match the distribution of the target domain which aids in training.

Nowadays, there are some researches applying domain adaptation to SAR-ATR. Huang et al. [18] discussed what, where, and how to transfer in SAR target classification and proposed a transitive transfer method via multisource data with domain adaptation. Wang et al. [19] proposed a model integrating meta-learning and adversarial domain adaptation to address domain shift and task transfer problems while a large amount of simulated data and only a small amount of annotated real samples are available. An unsupervised multilevel domain adaptation method based on domain adversarial learning is proposed in [20] to solve the problem of multiband SAR image classification. These previous studies indicate that it is of great application prospects to use domain adaptation in SAR target recognition with limited sample annotations, but how to transfer the positive information from the simulated data to the classification task of measured images is still a problem to solve.

On the other hand, more and more attention has been paid to the explainability of deep learning in recent years [21]. Explainability analysis aims to explore the features learned by the CNNs and to explain the decision basis of the trained models, so as to help better evaluate whether the decision mechanism is reliable. Undoubtedly, the reliability of the model is very important for SAR target classification. However, due to the unique imaging mechanism, the target and the background are mostly presented in the form of discrete scatter points in SAR images. As a result, the separability between the target and the background, between different types of target, and between different structures of the target is not strong. Therefore, the improvement of classification accuracy alone cannot guarantee the credibility of the model, and the decision-making mechanism of CNNs needs to be explored. Although some studies have focused on explainable deep learning and proposed corresponding methods [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], researches about explainable SAR-ATR based on CNNs are still very scarce [33], [34]. What is more, the current researches in the simulation-aided SAR target classification paid more attention to whether the information transfer between the two domains is successful. The analysis of changes in the extracted features before and after the transfer is insufficient, so the reliability and credibility of the model cannot be guaranteed.

To solve all these above problems, we develop a new network that integrates image reconstruction and subdomain alignment (SA), effectively improving the accuracy of simulation-aided SAR target classification with only a small number of measured sample annotations. Our main contributions can be summarized as follows.

1) We propose a novel model to achieve highly accurate simulation-assisted SAR target classification, using only a limited number of real sample annotations.
2) We adopt a domain-shared encoder and two domain-specific decoders to reconstruct samples from both the simulated and real domains. This approach allows the encoder to learn feature extraction without the use of labels. In addition, we make an attempt on using local maximum mean discrepancy (LMMD) to effectively use the pseudolabels generated during training to narrow the distribution of features of the same categories in different domains.
3) Explainability analysis methods have been used to identify the decision-making basis of the proposed model, which is very rare in other simulation-assisted SAR target recognition works. The results of the analysis provide insight into the main reasons for the performance improvement of the proposed model, thereby ensuring its reliability. Furthermore, the findings suggest that the proposed model places greater emphasis on the target and its shadow in the input images to generate more precise predictions, potentially having enlightening implications for future research in this field.

The rest of this article is organized as follows. In Section II, we briefly present the related work. Section III introduces the methods in detail. The experimental data and results are depicted in Section IV. The model decision analysis is given in Section V, and Section VI gives our conclusion.

II. Related Work

In this article, we are concerned about simulation-aided SAR-ATR and explainable SAR-ATR. Consequently, we will introduce the development of SAR-ATR and simulation-assisted SAR-ATR technology followed by several related studies on explainable SAR-ATR.

A. SAR-ATR Based on CNNs

Due to the excellent image classification performance and powerful feature extraction ability, CNNs have attracted much attention in SAR-ATR. Shao et al. [35] analyzed and compared the performance of a variety of representative CNN models on the SAR target classification task according to the accuracy rate, the number of parameters, training time, and other indicators. The results provided data support and effective reference for further model selection, combination, and optimization. Chen et al. [1] proposed all-convolutional networks (A-ConvNets), which used convolutional layers instead of fully connected layers to complete high-dimensional feature classification, reduced the parameters that need to be trained, and effectively alleviated overfitting.

Considering the significant differences between SAR images and optical images, many studies have tried to combine the characteristics of SAR data and neural networks to build a new SAR target recognition model. Huang et al. [36] proposed

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Deep SAR-Net, a dedicated deep learning framework for complex-valued SAR images, which can learn more valuable information in SAR images from both the spatial and frequency domains. The feature fusion framework (FEC) proposed by Zhang et al. [37] effectively improved the robustness of target recognition by constructing a visual word bag with the attribute scattering center feature parameters and fusing it with the features extracted from the magnitude images by CNN.

In addition, due to the difficulty in interpreting SAR images and the scarcity of large-scale and high-quality annotated datasets, related researches in recent years have begun to focus on SAR target recognition with few-shot samples. Pan et al. [38] proposed a metric-learning-based Siamese network to increase the amount of training data by sampling image pairs rather than single image input. Fu et al. [39] proposed a meta-learning framework named MSAR for the problem of few-shot SAR target recognition and studied the feasibility of using meta-learning to solve the problem of few-shot SAR target classification for the first time. Wang et al. [40] proposed a hybrid inference network (HIN) for few-shot SAR target recognition. In the first stage, the SAR images were mapped into an embedding space with an embedding network. In the second stage, the inductive inference and the transductive inference were adopted as the hybrid strategy to classify the samples in the embedding space. These algorithms emphasized how to dig the representation potential of few-shot labeled SAR samples, but ignored the possibility of representation transfer learning from other domains, such as the simulated SAR image domain.

B. Simulation-Aided SAR Target Classification

With the continuous development of SAR image simulation technology, SAR simulation data have the advantages of low cost and fast speed. Large-scale simulation SAR datasets with accurate labels provide a new possible way to achieve SAR target classification in the case of insufficient sample annotations. Therefore, some scholars began to explore the feasibility of simulation-assisted SAR target recognition. The classical SAR image simulation method calculates the radar cross section (RCS) of the SAR target to establish a 3-D model of the target and obtain a simulated image that is similar to the real image of the SAR target [41]. However, such methods have a strong dependence on the accuracy of the model and the veracity of parameter settings.

With the increasing attention of deep-learning-based image generation methods, generative adversarial network (GAN) [42] has been gradually applied to SAR image simulation tasks. Guo et al. [43] developed an end-to-end model that can generate simulated images based on real SAR images and proposed a clutter normalization method, which effectively avoids the network from falling into oversimulation of noise and improves the quality of the generated images. Gao et al. [44] constructed a jointly trained model with two discriminators by doubling the discriminator of the deep convolutional GAN (DCGAN). With the aid of generated high-quality images, their model improved the performance of semisupervised learning SAR target recognition networks. Niu et al. [45] proposed a framework that used CNN to obtain the mapping relationship between the simulation parameters and SAR images and realized intelligent extraction of simulation parameters based on SAR images. This method effectively reduced the difficulty in setting simulation parameters. At the same time, the advantages of electromagnetic simulation SAR images with reliable physical knowledge are retained, and the similarity of SAR simulation images is improved. In addition, Niu et al. [46] embedded CNN into the SAR target image simulation framework, so that the CNN can directly learn and fit the electromagnetic reflection models from real SAR images, which not only ensures the correctness and versatility of the simulation method but also significantly improves the fidelity of the simulated image. Although in some cases simulated images can be visually indistinguishable from real images, it is difficult for CNNs trained with simulated images to directly and accurately classify real samples [4].

Some relevant studies have shown that SAR simulation samples can provide effective information for SAR target classification. Malmgren-Hansen et al. [3] first discussed the feasibility of using simulation images for transfer learning to achieve real SAR target classification. The results show that when the real samples are insufficient, the transfer learning strategy of using simulation data for pretraining and fine-tuning with a few real data can effectively accelerate model convergence. Liu et al. [47] proposed a method to transfer simulated samples to be more similar to real samples in the image domain using cycle adversarial network (CycleGAN) [17], which can release more potential of simulated samples. Zhang et al. [48] proposed a two-stream deep network to extract the features of the entire image and image patches for more effective use of the SAR domain knowledge such as azimuth, amplitude, and phase information. Du et al. [49] proposed an adversarial encoding network (AEN) to extract common features between the simulated data and real data to improve the classification performance in the case of insufficient labeled real data. He et al. [50] proposed a task-driven domain adaptation (TDDA) transfer learning method to introduce SAR imaging condition information into the training stage to alleviate the degradation of classification performance caused by the differences between training and testing data, but still requires a large number of annotated samples. Song et al. [4] proposed a method to suppress the influence of the distinct but nonessential texture that only existed in simulated images, which performed a series of preprocessing, including speckle suppression, style adjustment, and background deletion on the inputs, and then input them to the neural network, realizing the simulation-assisted zero-shot SAR target recognition. However, their experiments only replace one category among the ten types of targets in the measured dataset with the simulated images, which is limited to verifying the effect of the simulated images on supporting the real image classification. In addition, the above studies have paid little attention to the internal decision-making mechanism of the trained models. Some problems, such as why the simulated images cannot directly support the classification of the measured samples, and why the proposed models can

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improve the performance of the simulation-aided SAR target classification, cannot be explained clearly, so it is necessary to explore the explainability of the network.

C. Explainable SAR-ATR

The lack of explainability of deep learning limits its development and application to a certain extent, and methods with both performance and explainability are more needed in many aspects. The neural network itself is not totally a black box. Its entire training process and specific values of parameters are known. The black-box property is mainly due to the highly nonlinear characteristic of the neural networks, which makes it impossible to intuitively understand the specific meaning and behavior.

In recent years, there have been studies to explain the internal decision-making mechanism of neural networks from different views. The existing deep learning explainability methods are mainly divided into three types: 1) hidden layer analysis methods [22], [23], [24] can qualitatively analyze hidden features through visualization; 2) model substitution methods [28], [29] improve explainability using low-complexity models as substitutions; and 3) localization explanation methods [26], [27], [32] aim to find the feature area of a certain category to explain the classification results.

In the field of SAR-ATR, explainability is also a topic of great concern. Belloni et al. [33] proposed a set of analytical tools to compare the influence of different regions of the input on CNN's decision-making and find that the most important areas for the CNN are often located on specific parts of the target. In [34], local interpretable model-agnostic explanations (LIMEs) [32] are used to provide explanations for the trained CNN model to justify the classification results. These works inspired us to explore the changes in the decision-making basis of the model to evaluate whether the performance improvement is reliable.

III. Methods

In this section, we will first introduce the specific architecture and the pipeline of the proposed model. Then the main structures, the image reconstruction module and the SA module, are described in detail. The loss function is listed subsequently. The explainability analysis is described at the end.

A. Overall Architecture

The overall framework of our model is shown in Fig. 1, which contains three stages: training, testing, and explaining. Fig. 1(a) presents the training pipeline of our model. The model is mainly composed of the shared encoder and classifier, the simulation domain decoder, the real domain decoder, and the SA module. The encoder and the classifier jointly complete the classification of the input simulated or measured images. The encoder and the decoder realize the image reconstruction of the corresponding domain of the decoder. The SA module takes the sample labels or pseudolabels and the features as inputs and measures the distance between the features of the same category from different domains according to the (pseudo) labels to achieve SA. In Fig. 1(a), the gray parts correspond to the domain-shared structures, including the encoder, the classifier, and the SA module. The blue and orange parts represent the domain-specific structures, in which blue belongs to the simulated domain and orange belongs to the real domain.

During the training procedure, the input data include three types: a large number of labeled simulation samples $D_s = \{(x_i^s, y_i^s)_{i=1}^n\}$, a small number of labeled real samples $D_r = \{(x_i^r, y_i^r)_{i=1}^l\}$, and a large amount of unlabeled real samples $D_{r'} = \{(x_i^{r'})_{i=1}^m\}$, where s represents the simulated domain, r represents the labeled real domain, r' represents the unlabeled real domain, x_i^s indicates the ith sample in the simulated domain, and y_i^s corresponds to the label of the ith sample in the simulated domain. It should be noted that $y_i^s \in \{0, 1, 2, \ldots, N-1\}$, where N is the number of categories. What is more, we use θ_e, θ_c, θ_d^s, and θ_d^r to represent the parameters of the encoder, classifier, simulated domain decoder, and real domain decoder, respectively, and $G_e(\cdot)$, $G_c(\cdot)$, $G_d^s(\cdot)$, and $G_d^r(\cdot)$ represent the corresponding mapping functions. Then for the input x, there are feature vectors $f = G_e(x; \theta_e)$, the prediction label $\hat{y} = G_c(f; \theta_c)$, and the reconstructed image $\hat{x} = G_d(f; \theta_d)$.

As illustrated in Fig. 1(b), upon completion of the training phase, the samples requiring classification are input into the classification network, which consists of an encoder and a classifier, to obtain prediction results. For model explanation purposes, we use an instance-level strategy. Specifically, a single sample is input into the network to generate a prediction vector, and explainability analysis methods are subsequently executed to produce heat maps that reflect the decision-making basis of the network. This process is depicted in Fig. 1(c) and will be elaborated upon in further detail later.

B. Image Reconstruction

In practical scenarios, it is not difficult to obtain a large amount of unlabeled SAR images but to interpret and annotate them. Therefore, how to perform feature extraction without relying on data labels is crucial to improve the accuracy of SAR target classification. Deep autoencoder networks [51] can achieve feature dimension reduction by encoding and compressing the input first, and then decoding and reconstructing it. In other words, deep autoencoder networks can learn feature extraction without relying on annotations. Ghifary et al. [14] also emphasized the impact of image reconstruction in the target domain on improving the domain adaptation ability of the network. Inspired by these works, in our model, the encoder and the domain-specific decoders are designed to reconstruct the images from the corresponding domain to help the encoder learn an effective feature extraction method without supervised learning. As demonstrated in Tables I and II, the decoder and the encoder adopt symmetrical structures. That is, the encoder consists of five convolutional layers, while the decoder is composed of five deconvolutional layers. It should be noted that the notation Conv(c, k, s, p) refers to the output channel being c, the kernel size being k, the stride being s, and the padding being p. The same notation applied to

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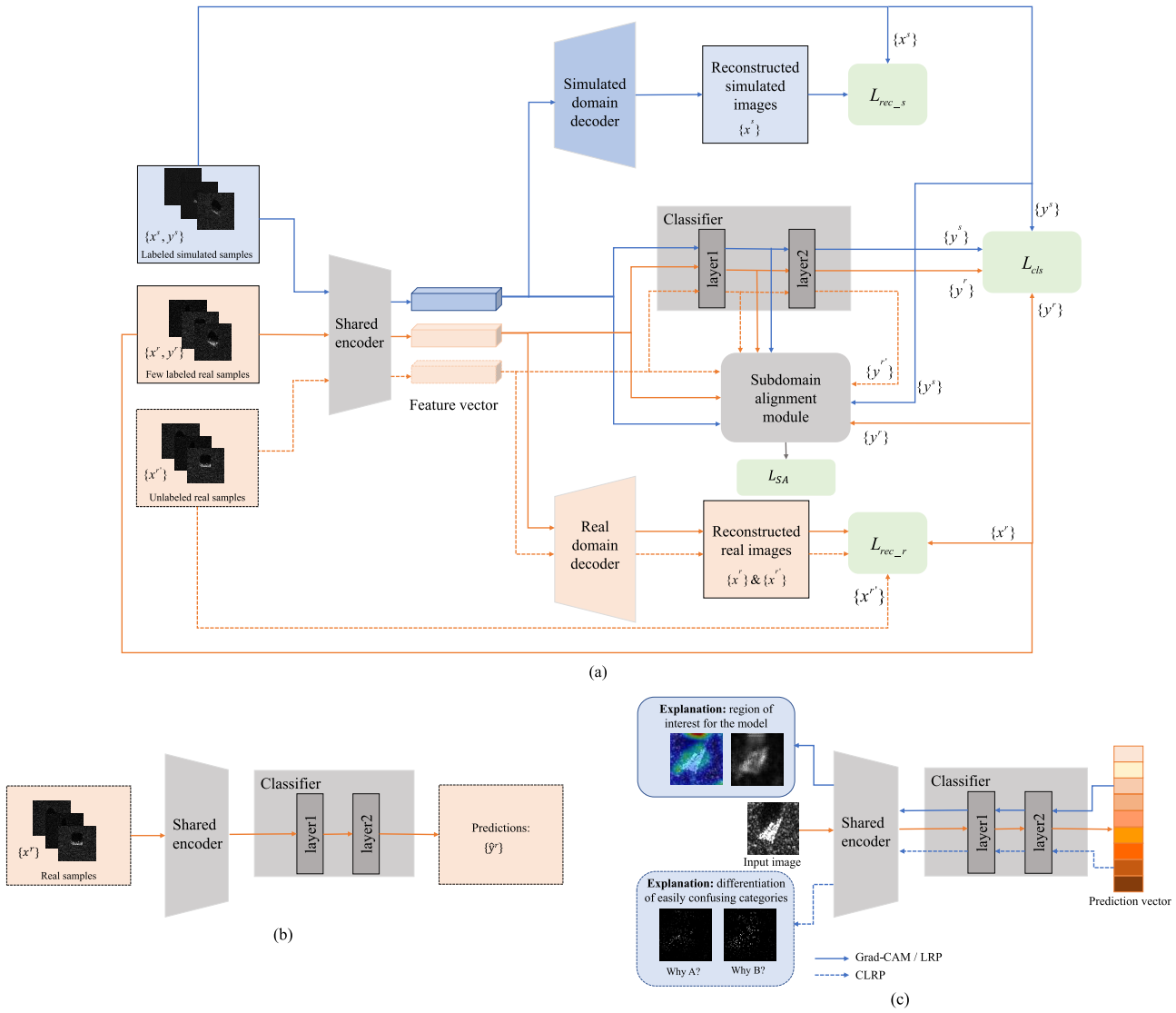


Fig. 1. Pipeline of the proposed method. (a) Model training. (b) Model testing. (c) Model explaining. A and B refer to categories.

DeConv(c, k, s, p). As depicted in Fig. 1, the encoder is domain-shared, while the decoders are domain-specific. The structure of both the simulated and real domain decoders is identical; however, their weights are determined by their respective training processes. The optimization objective of the encoder–decoder network is to minimize the reconstruction loss, so as to help the encoder extract valid features by improving the authenticity of the reconstructed images. The commonly used mean square error (mse) loss is used as the reconstruction loss function. The calculation of L_{rec} can be expressed as

$$L_{\text{rec}} = L_{\text{rec}_s} + L_{\text{rec}_r} \tag{1}$$

where L_{rec_s} and L_{rec_r} are calculated as follows. The minimization of L_{rec_s} and L_{rec_r} optimizes the simulated domain decoder and the real domain decoder, respectively, and jointly influences the domain-shared decoder

$$L_{\text{rec}_s} = \frac{1}{n}\sum_{i=1}^{n}\left\|\widehat{x_i^s} - x_i^s\right\|_2^2 \tag{2}$$

TABLE I
DETAILED ARCHITECTURE OF THE ENCODER

Layer	Structure	The Shape of Output
Layer 1	Conv(32, 6, 2, 2) ReLU	$32 \times 32 \times 32$
Layer 2	Conv(64, 6, 2, 2) ReLU	$16 \times 16 \times 64$
Layer 3	Conv(128, 4, 2, 1) ReLU	$8 \times 8 \times 128$
Layer 4	Conv(256, 4, 2, 1) ReLU	$4 \times 4 \times 256$
Layer 5	Conv(512, 4, 1, 0) ReLU	$1 \times 1 \times 512$

$$L_{\text{rec}_r} = \frac{1}{l}\sum_{i=1}^{l} l\left\|\widehat{x_i^r} - x_i^r\right\|_2^2 + \frac{1}{m}\sum_{i=1}^{m}\left\|\widehat{x_i^{r\prime}} - x_i^{r\prime}\right\|_2^2 \tag{3}$$

where x_i^* represents the ith sample of the specified domain, $\widehat{x_i^*}$ is the corresponding reconstruction result, and n, l, and m are the number of labeled simulated samples, labeled real samples, and unlabeled real samples, respectively.

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TABLE II
DETAILED ARCHITECTURE OF THE DECODER

Layer	Structure	The Shape of Output
Layer 1	DeConv(256, 4, 1, 0) ReLU	$4 \times 4 \times 256$
Layer 2	DeConv(128, 4, 2, 1) ReLU	$8 \times 8 \times 128$
Layer 3	DeConv(64, 4, 2, 1) ReLU	$16 \times 16 \times 64$
Layer 4	DeConv(32, 6, 2, 2) ReLU	$32 \times 32 \times 32$
Layer 5	DeConv(1, 6, 2, 2) Tanh	$64 \times 64 \times 1$

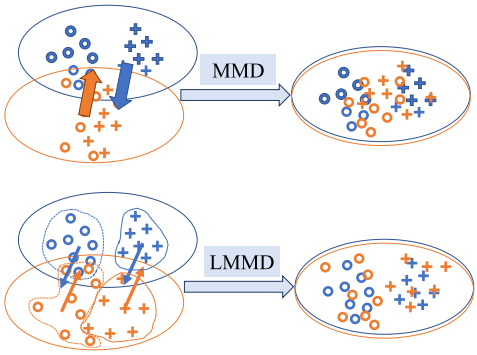


Fig. 2. MMD versus LMMD while performing domain adaptation.

C. Subdomain Alignment

LMMD is first proposed in [8]. As shown in Fig. 2, unlike maximum mean discrepancy (MMD) [52], which only considers the domain attributes, LMMD also considers the category of the inputs and then divides them into different subdomains to perform SA. We use s, r, and r' to represent the distribution of labeled simulation data, labeled real data, and unlabeled real data, respectively. Then the LMMD between s and r is defined in the following equation, and its unbiased estimation is calculated as (5):

$$d_k^2(s,r) \triangleq E_c \| E_{s^c}[\phi(x^s)] - E_{r^c}[\phi(x^r)] \|_{H_k}^2 \tag{4}$$

where x^s and x^r are the instances of simulated domain D_s and real domain D_r, respectively, s^c and r^c are the distribution of D_s^c and D_r^c, respectively, which are exactly the aforementioned subdomains, and D_*^c means the sample set of the cth category. Here, H_k means the reproducing kernel Hillbert space (RKHS) endowed with a characteristic kernel k, $\phi()$ denotes the feature mapping that maps the original samples to RKHS, and the kernel k means $k(x^s, x^r) = \langle \phi(x^s), \phi(x^r) \rangle$

$$d_k^2(s,r) = \frac{1}{N} \sum_{c=1}^{N} \left\| \sum_{x_i^s \in D_s} w_i^{sc} \phi\left(x_i^s\right) - \sum_{x_j^r \in D_r} w_j^{rc} \phi\left(x_j^r\right) \right\|_{H_k}^2 \tag{5}$$

where N is the number of the categories, and w_i^{sc} and w_j^{rc} denote the weight of x_i^s and x_j^r belonging to the category c. It should be noted that $\sum_i w_i^{sc} = \sum_j w_j^{rc} = 1$, and w_i^c can be calculated as

$$w_i^c = \frac{y_{ic}}{\sum_{(x_j, y_j) \in D} y_{jc}} \tag{6}$$

where y_{ic} is the cth value in the vector $y_i = [y_{i1}, y_{i2}, \ldots, y_{iN}]^T$, denoting the probability of the ith sample belongs to the cth class. For the labeled samples, y_i is the one-hot vector of its real class.

Considering that $k(x^s, x^r) = \langle \phi(x^s), \phi(x^r) \rangle$, the empirical estimate of (5) can be expressed as

$$\begin{aligned} d_k^2(s,r) = \frac{1}{N} \sum_{c=1}^{N} \Bigg[& \sum_{i=1}^{n} \sum_{j=1}^{n} w_i^{sc} w_j^{sc} k\left(x_i^s, x_j^s\right) \\ & + \sum_{i=1}^{l} \sum_{j=l}^{l} w_i^{rc} w_j^{rc} k\left(x_i^r, x_j^r\right) \\ & - 2 \sum_{i=1}^{n} \sum_{j=1}^{l} w_i^{sc} w_j^{rc} k\left(x_i^s, x_j^r\right) \Bigg]. \end{aligned} \tag{7}$$

Note that (7) shows how to calculate the LMMD between the labeled simulation data and the labeled real data, and the LMMD between the labeled simulation data and the unlabeled real data $d_k^2(s, r')$ can also be calculated like this. While for the unlabeled real data $D_{r'}$, we use the predicted results which are activated by softmax as a substitute to calculate $w_j^{r'c}$ as (6). The predicted vector $\hat{y}_i^r$ largely reflects the probability that x_i^r belongs to each class. So the optimization objective of SA is to minimize L_{SA}, which can be expressed as

$$L_{\mathrm{SA}} = \sum_{q=l_1}^{l_2} d_k^2\left(D_s^q, D_r^q\right) + \mathbb{I}(\mathrm{iter} > \mathrm{iter}_{\mathrm{SA}}) \gamma \sum_{q=l_1}^{l_2} d_k^2\left(D_s^q, D_{r'}^q\right) \tag{8}$$

where l_1 and l_2 are the start and end layers between which LMMD is calculated, respectively, D_*^q denotes the hidden representations of the qth layer, γ is the hyperparameter that adjusts the weight of $d_k^2(D_s^q, D_{r'}^q)$, and $\mathbb{I}(\mathrm{iter} > \mathrm{iter}_{\mathrm{SA}})$ is the indicator function which equals 1 if iter $>$ iter$_{\mathrm{SA}}$, else equals 0.

D. Classifier

After feeding the features output by the encoder, the classifier gives the predictions. As shown in Table III, the classifier in our model no longer uses the conventional fully connected layer but uses convolutional layers instead, which can take advantage of weight sharing to effectively reduce the number of parameters and alleviate overfitting to a certain extent. The classification loss L_{cls} is calculated as

$$\begin{aligned} L_{\mathrm{cls}} = & \frac{1}{n} \sum_{i=1}^{n} J_y\left(G_c\left(G_e\left(x_i^s; \theta_e\right); \theta_c\right), y_i^s\right) \\ & + \frac{1}{l} \sum_{j=1}^{l} J_y\left(G_c\left(G_e\left(x_j^r; \theta_e\right); \theta_c\right), y_j^r\right) \end{aligned} \tag{9}$$

where n and l are the number of the labeled simulated and real images, respectively, and $J_y(\cdot)$ is the cross-entropy loss function.

E. Loss

To sum up, the optimization objective of the proposed model can be divided into three parts: the classification loss L_{cls}, the

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TABLE III
DETAILED ARCHITECTURE OF THE CLASSIFIER

Layer	Structure	The Shape of Output
Layer 1	Conv(256, 1, 1, 0) ReLU	$1 \times 1 \times 256$
Layer 2	Conv(9, 1, 1, 0)	$1 \times 1 \times 9$

reconstruction loss L_{rec}, and the SA loss L_{SA}. The total loss L_{all} can be expressed as

$$L_{\text{all}} = L_{\text{cls}} + \lambda_{\text{rec}} \cdot L_{\text{rec}} + \lambda_{\text{SA}} \cdot L_{\text{SA}} \tag{10}$$

where λ_* is the hyperparameter that adjusts the weight of L_*.

F. Explainability Analysis

We hope to verify the reliability of the proposed method by comparing and analyzing the decision-making basis before and after its implementation. To achieve this, we primarily use the localization explanation methods, namely, gradient-weighted class activation mapping (Grad-CAM) [27], layerwise relevance propagation (LRP) [30], and contrastive LRP (CLRP) [31]. Importantly, these three methods can be implemented without modifying the original neural network. Specifically, Grad-CAM uses the gradients of any target category, flowing into the final convolutional layer, which is the last layer of the encoder in our experiment, to produce a coarse localization map reflecting the influence of different image regions on category prediction. To interpret neural networks, LRP assigns the classification decision into contributions of its input elements by decomposing the relevance of the jth neuron in the $L+1$th layer into all the neurons in the Lth layer. However, this decomposition operation is solely satisfied at the last layer of neural networks, leading to instance-specific rather than category-discriminative explanations, mainly reflecting the input elements that significantly impact decision-making. By constructing a dual virtual concept to eliminate information unrelated to the target category from the LRP explanations, CLRP can provide category-discriminative explanations.

Our focus when explaining network decisions is primarily on the encoder and the classifier. As shown in Fig. 1(c), Grad-CAM and LRP are initially used to locate where the network is paying attention, and to interpret the semantic meaning of these areas. Subsequently, CLRP is adopted to obtain category-discriminative explanations, and the reasons for the classification performance improvement are analyzed based on the misclassified samples. In addition to the aforementioned works, we supplement the model explanation by observing changes in feature distribution and the reconstructed images by the decoders, which reflect the extracted information of the encoder.

IV. EXPERIMENT

To evaluate the effectiveness of the proposed dual-branch image reconstruction and SA (DBRSA), comprehensive experiments have been conducted in this section. First, the simulated and real datasets used in this article are introduced. Next, experimental settings and implementation details are described. Following that, the experimental results are shown to confirm the effectiveness of our model. Finally, the performance of our model is analyzed by ablation experiments.

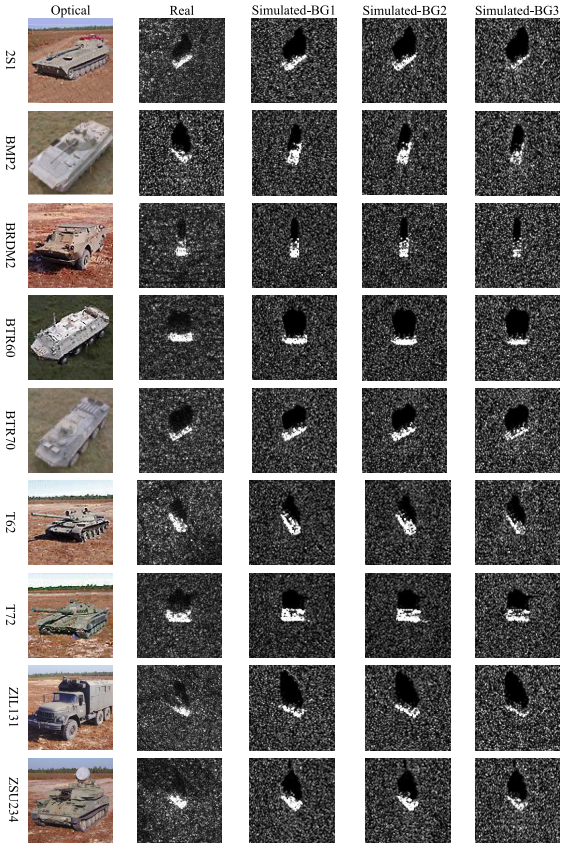


Fig. 3. Optical images of the nine types of targets and the corresponding simulated and real SAR images under typical azimuths.

A. Dataset

To validate the effectiveness of DBRSA, the moving and stationary target acquisition and recognition (MSTAR) dataset is used in our work as the real data, which has been widely used in SAR-ATR studies. The simulated data are provided by the State Key Laboratory of Millimeter Waves of Southeast University. As illustrated in Fig. 3, the target categories in our datasets include 2S1, BMP2, BRDM2, BTR60, BTR70, T62, T72, ZIL131, and ZSU234. The number of real samples in our experiment is shown in Table IV. The labeled real dataset D_r is obtained by randomly sampling k samples from the real data with a depression angle of 17°, where $k \in \{10, 30, 50\}$. Among them, the remaining samples are taken as the unlabeled real dataset $D_{r'}$. The accuracy of the model is validated on the real samples under 15°. To ensure diversity, the simulated images include two depression angles (15° and 17°) and three kinds of backgrounds: the grassland automatically generated by the simulation software (recorded as BG-1), the grassland generated by manually setting parameters with reference to the optical images (recorded as BG-2), and the sandland generated by manually setting parameters with reference to optical images (recorded as BG-3). For each category and each background, the simulated images are generated every 1° azimuth, with a total of 19 494 images.

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TABLE IV
SIZE OF THE REAL SAR DATASET

Category	Type	Depression Angle	Number
2S1	B01	17°	299
		15°	274
BMP2	9563	17°	233
		15°	195
BRDM2	E-71	17°	298
		15°	274
BTR60	7532	17°	256
		15°	195
BTR70	C71	17°	233
		15°	196
T62	A51	17°	299
		15°	273
T72	132	17°	232
		15°	196
ZIL131	E12	17°	299
		15°	274
ZSU234	D08	17°	299
		15°	274

TABLE V
PARAMETERS FOR THE MODEL TRAINING PROCEDURE

Parameter	Value
batch size	64
optimizer	Adam
lr_{cls}	0.001
lr_{rec}	0.005
λ_{rec}	10
λ_{SA}	0.5
$iter_{max}$	30000
$iter_{SA}$	2500

B. Experimental Settings

In the experiments, all the input images are center cropped to 64×64. The specific parameter settings are listed in Table V, where iter_{SA} refers to that $\gamma = 0.5$ in (8) when the current training iteration exceeds this value, otherwise $\gamma = 0$; lr_{cls} is the learning rate of the encoder and the classifier; and lr_{rec} is the learning rate of the decoder. Our experiments were carried out in PyTorch 1.4.0, Linux3.10.0. The hardware is based on Intel Xeon[1] Gold5218R CPU at 2.10-GHz CPU and NVIDIA Tesla V100S GPU.

C. Results

Fig. 4 shows the confusion matrixes of the baseline network and the proposed DBRSA on the MSTAR test set under different training set settings. It can be proved that DBRSA has significantly improved the recognition accuracy of each category, and the model's ability to identify similar targets, such as BTR60 and BTR70, is also better. What is more, Fig. 5 and Table VI present the performance comparison of the proposed model and other existing models on the dataset used in this article. We use k-shot to indicate that there are k labels of real samples in the training set in the following text, and $k \in \{10, 30, 50\}$. The relevant descriptions of the mentioned models are as follows.

[1]Registered trademark.

TABLE VI
ACCURACY OF DIFFERENT METHODS ON OUR DATASET

Methods	10 shot	30 shot	50 shot
Baseline	0.4040	0.6216	0.7406
DAN [6]	0.4995	0.6581	0.7859
DRCN [14]	0.4296	0.6722	0.8565
TLNet [3]	0.3919	0.6267	0.7657
Method in [53]	0.5462	0.7071	0.8145
AEN [49]	0.5496	0.8042	0.8814
DBRSA(ours)	0.7736	0.9242	0.9758

1) *Baseline:* The model is only trained with the optimization objective of L_{cls}.
2) *DAN:* The proposed method in [6]. It uses multikernel MMD (MK-MMD) to measure the discrepancy of features from different domains.
3) *DRCN:* The proposed method in [14]. A shared encoder is trained for two tasks: supervised classification of simulated data and unsupervised reconstruction of real data to ensure the learned representations not only preserve discriminability but also extract useful information from the real data.
4) *TLNet:* The proposed method in [3]. This model is first trained on simulated data and then fine-tuned with the labeled real data.
5) *Method in [53]:* This method integrates MK-MMD and domain adversarial training to solve the domain shift problem during the task transfer from simulated SAR image classification to the real one.
6) *AEN:* The proposed model in [49]. This model uses one decoder to reconstruct the simulated and real images and learns domain-invariant features through adversarial training between the encoder and the discriminator.

It turns out that our method achieves optimal performance under three real subsets in this article. We suspect that due to different categories and quantities of simulation data, and the randomness of the sampling of real subsets, transfer learning network (TLNet) can hardly improve the recognition accuracy under the sparse real subset, which is slightly different from their results. Deep adaptation network (DAN) [53] and AEN only consider the domain attributes but ignore the category attributes of features when performing feature alignment. Although it improves domain confusion, it may not enhance category discrimination significantly. Deep reconstruction-classification network (DRCN) only reconstructs the real data, but reconstructing simulated SAR data could also provide useful information about the target structure.

D. Ablation Studies

In this section, we have conducted multiple experiments to explore how the image reconstruction task and the SA module influence the performance of the model, so as to verify the effectiveness of our method.

1) Loss Function: To validate the effect of each part, the experiments are completed by changing the optimization objective, and the specific results are listed in Table VII.

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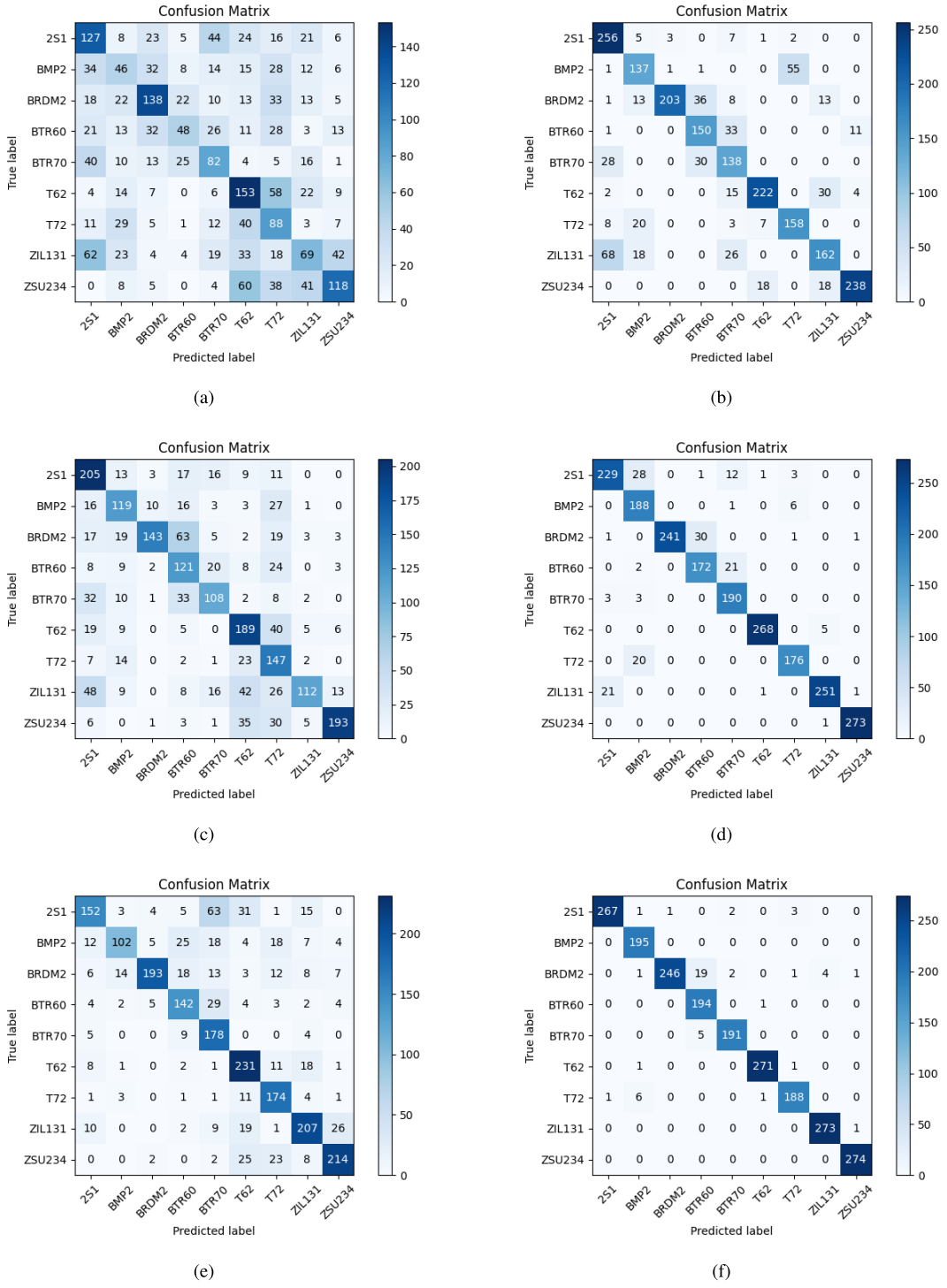


Fig. 4. Confusion matrixes of the baseline model and the proposed model. (a) Confusion matrix of the baseline network under ten-shot. (b) Confusion matrix of the proposed model under ten-shot. (c) Confusion matrix of the baseline network under 30-shot. (d) Confusion matrix of the proposed model under 30-shot. (e) Confusion matrix of the baseline network under 50-shot. (f) Confusion matrix of the proposed model under 50-shot.

The model whose optimization objective only includes L_{cls} is the baseline network in Section IV-C. It can be seen from Table VII that the reconstruction of real images can provide significant positive information for the classification task. As for the simulation images, it seems that the reconstruction task has relatively limited positive information provided, perhaps because there are a lot of corresponding labels during training. Compared with the case where the number of labeled real samples is large, L_{SA} can bring more obvious performance improvement when there are fewer labels.

The performance of L_{SA} depends on two important parameters: γ and iter_{SA}. The parameter γ controls the weight of the LMMD between the simulated samples and the pseudolabeled real samples. The parameter iter_{SA} determines when the pseu-

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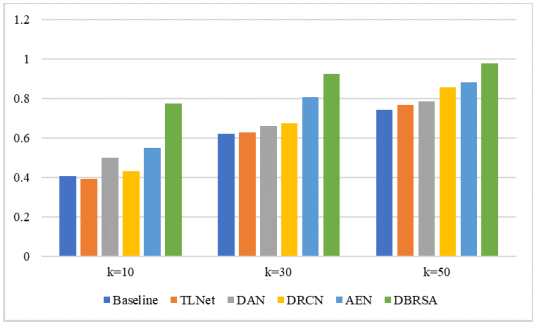
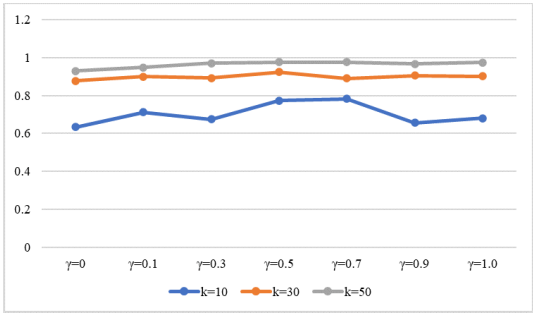


Fig. 5. Comparison of classification accuracy of different methods.

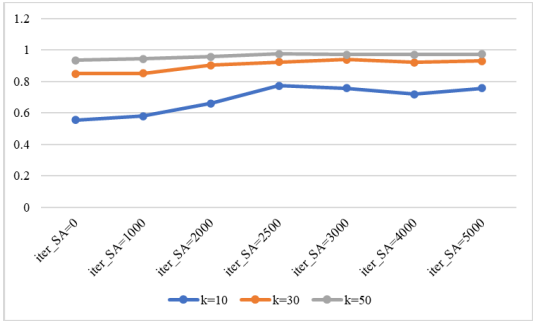
TABLE VII

RESULTS OF ABLATION EXPERIMENTS WITH DIFFERENT SUBSETS OF REAL SAR IMAGES

k shot	L_{cls}	L_{rec_r}	L_{rec_s}	L_{SA}	Accuracy
10	✓				0.4040
	✓	✓			0.6824
	✓	✓	✓		0.6875
	✓	✓	✓	✓	0.7736
30	✓				0.6215
	✓	✓			0.8539
	✓	✓	✓		0.8718
	✓	✓	✓	✓	0.9242
50	✓				0.7406
	✓	✓			0.9311
	✓	✓	✓		0.9577
	✓	✓	✓	✓	0.9758



(a)



(b)

Fig. 6. Classification accuracy of different parameters. (a) Classification accuracy of different γ. (b) Classification accuracy of different iter_{SA}.

dolabels are used in training. We investigate the influence of γ and iter_{SA} on the accuracy of our model in Fig. 6. We observe that when there are few labeled real samples, the quality of pseudolabels may be low, so a large value of γ may harm the performance. On the other hand, if $\gamma = 0$, ignoring the pseudolabels completely can also reduce the accuracy. Similarly, when iter_{SA} is too small, the model has not converged yet, so introducing pseudolabels at this stage may be detrimental. Based on our experiments, we recommend setting $\gamma = 0.5$ and $\text{iter}_{\text{SA}} = 2500$.

TABLE VIII

CLASSIFICATION ACCURACY OF DIFFERENT REAL SUBSETS UNDER DBR OR SBR

k shot	SBR	DBR
10	0.6962	0.7736
30	0.8943	0.9242
50	0.9646	0.9758

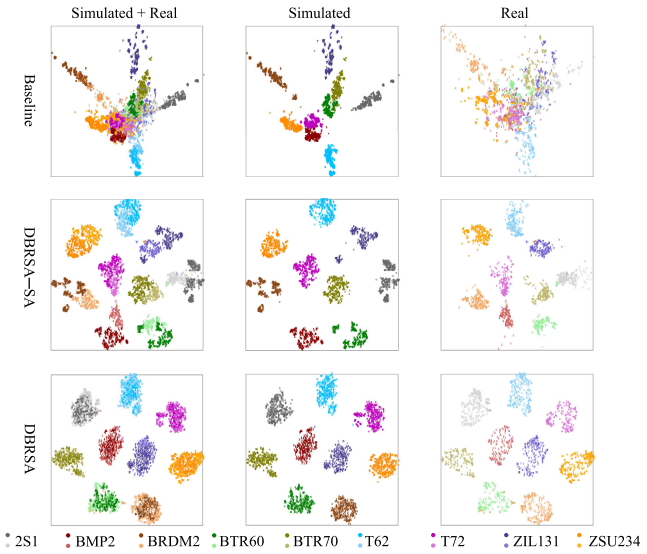


Fig. 7. Visualization of extracted features using t-SNE.

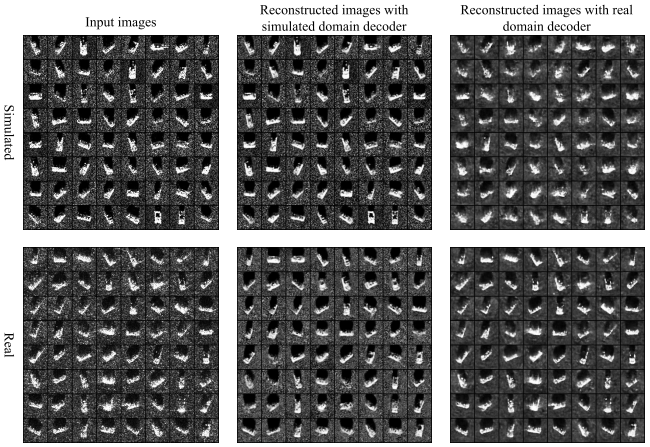
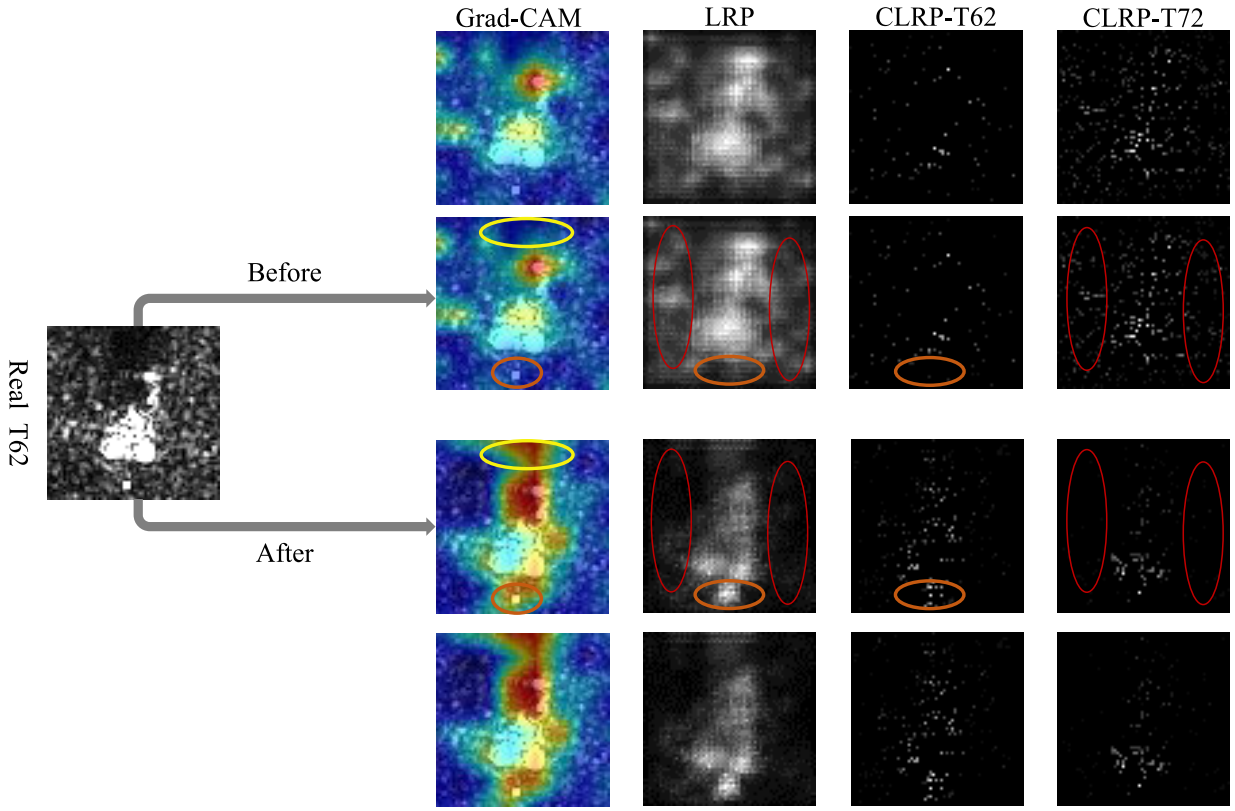


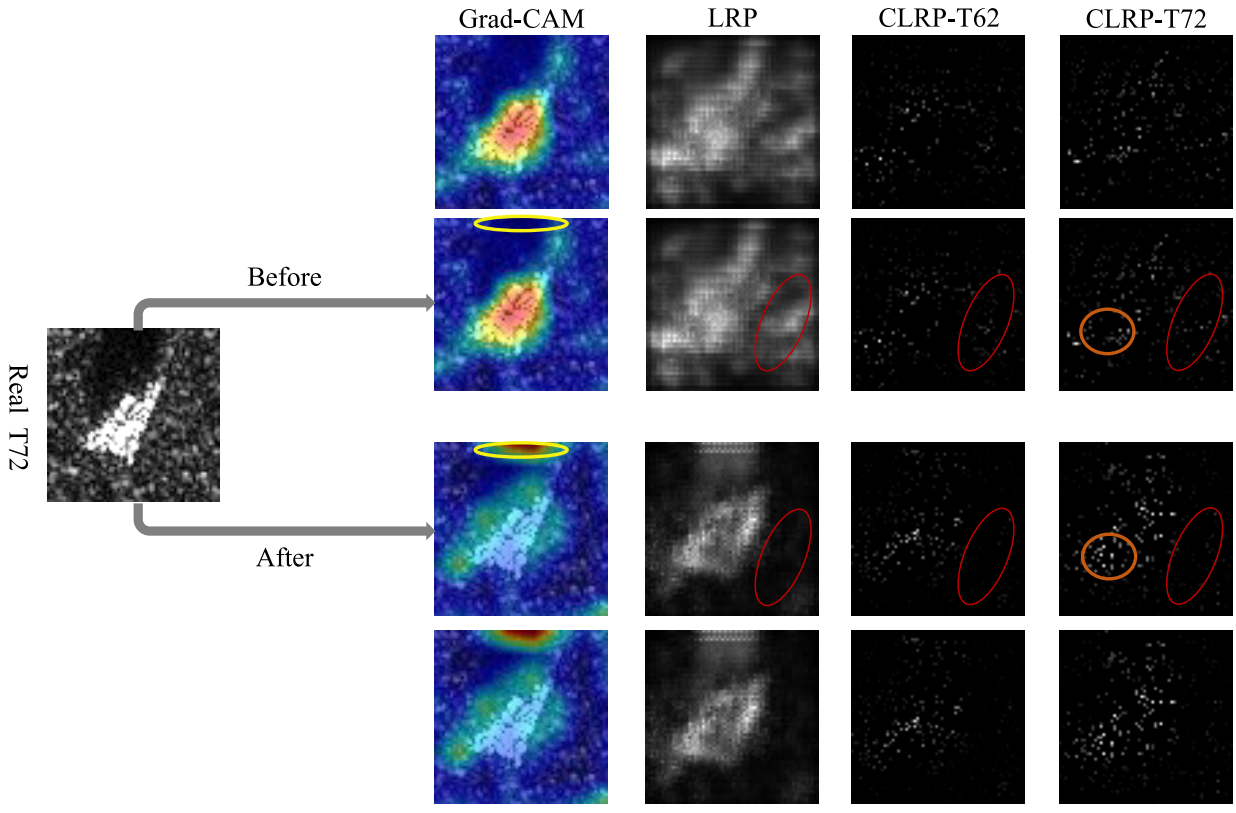
Fig. 8. Reconstructed results of simulated images and the real ones with the decoder of different domains.

2) DBR or SBR: In addition, we are concerned about whether the dual-branch reconstruction (DBR) structure is more meaningful than the single-branch (SBR) structure, i.e., reconstructing the simulated and real images with the same decoder. Therefore, we compared the classification results using DBR and SBR, as shown in Table VIII. It can be concluded that the less the real samples are labeled, the more obvious the improvement brought by DBR.

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(a)



(b)

Fig. 9. Decision basis visualization of the model trained without using DBRSA. (a) Input is a real image of T62, which was misjudged as T72 before using DBRSA and corrected after using it. (b) Input is a real image of T72, which was misjudged as T62 before using DBRSA and corrected after using it.

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V. Analysis

Taking the experiment under 50-shot real subset as an example, we visualize the extracted features of Layer 1 in the classifier by t-distributed stochastic neighbor embedding (t-SNE) [54]. As shown in Fig. 7, the similar colors represent the same category, the dark colors correspond to the simulated targets, the light colors refer to the real targets, and the numbers 0–8 denote 2S1, BMP2, BTR60, BTR70, T62, T72, ZIL131, and ZSU234, respectively. For the baseline model, which neither uses DBR nor performs SA, the simulated features can be clearly classified, but the real features cannot. After implementing DBR, the features of real samples become easier to classify. Furthermore, after using SA, the features of real samples are closer to the features of the same simulated category. It indicates that our method can effectively narrow the distribution of the simulated and real categories.

The reconstructed results of the input images from different domains using different domain decoders are presented in Fig. 8. Reconstructing the real images using the simulated domain decoder refers to inputting the real features encoded by the encoder to the simulated domain decoder. The same is true for reconstructing the simulated images using the real domain decoder. As can be seen from Fig. 8, the reconstructed images by the same decoder have similar image styles. And the simulated images have more supervision information during the training procedure, so the reconstructed results have relatively more details. In addition, compared with the original input images, there is no obvious distortion in the target contour, the relative position of the target and the shadow, the target azimuth, and other information when the simulated domain decoder is used to reconstruct the real images, and vice versa. This indicates that the features extracted by the encoder effectively retain the information.

It can be known from the confusion matrixes in Fig. 4(c) that there is a certain amount of mutual misjudgment between T62 and T72. To more intuitively explore the decision basis of the model, we use Grad-CAM [27], LRP [30], and CLRP [31] to visualize the influence of different regions/pixels of the inputs on the decisions. The results are exhibited in Fig. 9. Note that in Grad-CAM explanations, red regions correspond to high scores for the predicted class. While in LRP and CLRP explanations, white dots correspond to high scores for the predicted or specified class. The first and second rows correspond to the decision basis before using DBRSA, and the third and fourth rows correspond to those after using DBRSA. For the sake of understanding, we use elliptical boxes of different colors in the second and third rows to mark the positions with large differences in the decision basis before and after. As shown in the red boxes, the baseline model has a strong response to the background noise in the real images, while the DBRSA pays less attention to the background. What is more, the yellow boxes indicate that after using DBRSA, the shadows of the targets have an impact on the model's decision, which seems to help the model make a more accurate judgment. In addition, it could be concluded from the orange boxes that after introducing DBRSA, the model has a stronger response to the barrel of the target. To sum up, the model optimized by DBRSA has different focuses on T62 and T72, to better distinguish the two categories and improve the overall recognition accuracy.

VI. Conclusion

In this article, a novel framework combining image reconstruction and SA is proposed to solve the domain shift problem while performing simulation-aided SAR target classification. First, the reconstruction task helps the encoder extract more effective features while there are not enough annotations. Second, the SA module implemented by LMMD is proven to significantly narrow the feature distribution of similar targets in different domains. The effectiveness of the proposed method is verified by both comparative and ablation experiments. Third, to better improve the explainability of our model, we make a full comparative analysis of the feature distribution, encoded information, and decision basis, which ensures that the performance improvement is credible. It could be concluded that the encoder trained according to our method learns an effective feature representation method, which can retain target contour, azimuth, and other important information of inputs from different domains. The explanation generated by Grad-CAM, LRP, and CLRP demonstrated that the proposed model pays more attention to the target and its shadow, rather than the background.

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